



Internship Presentation

Development of Automated Web Scraping Systems and Internal Software Tools

University

IILM University

Student

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Enrollment No.

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Program

B.Tech CSE

Organization

EduSkills Academy

About the Internship

Program Details

- **Duration:** 10-week Machine Learning & Data Science program
- **Completion Date:** March 28, 2026
- **Certificate ID:** 2026-0018CCED6

Organization

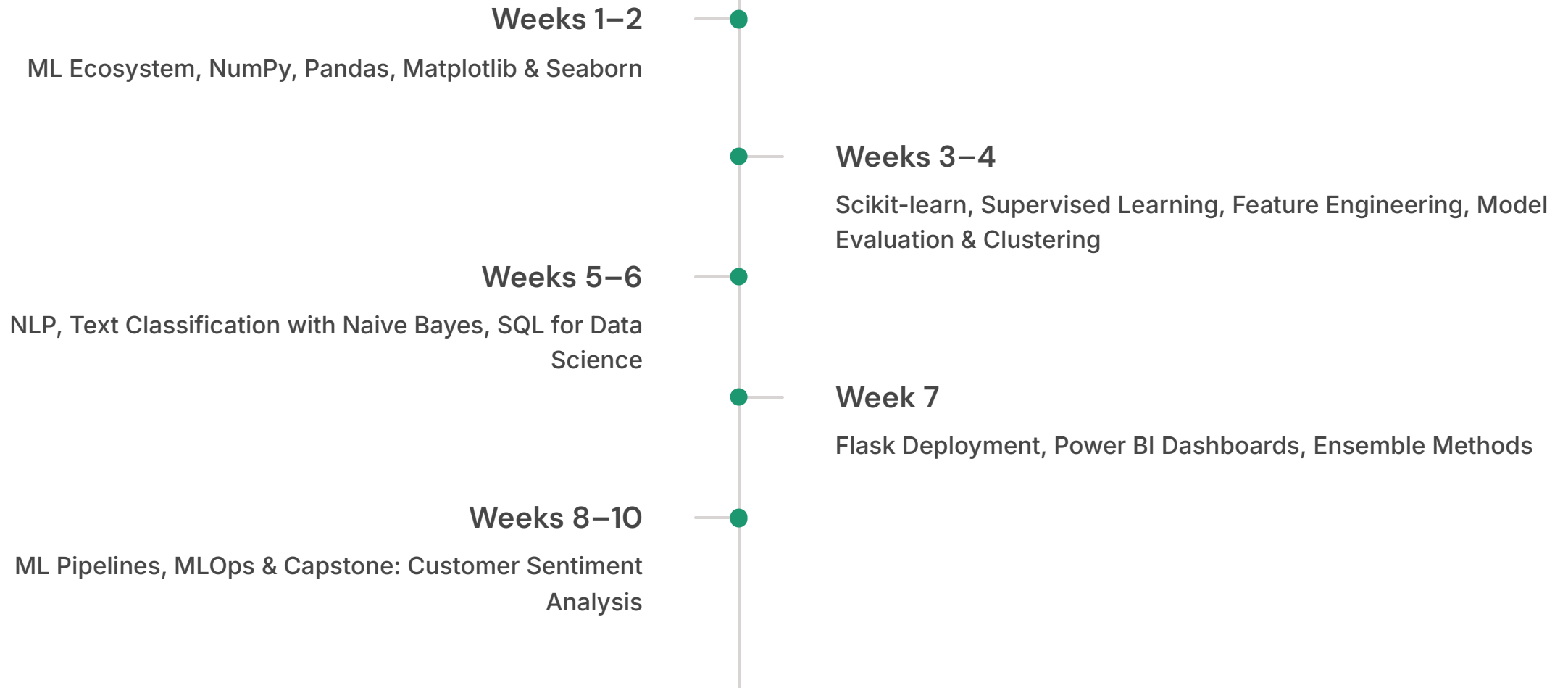
EduSkills Academy

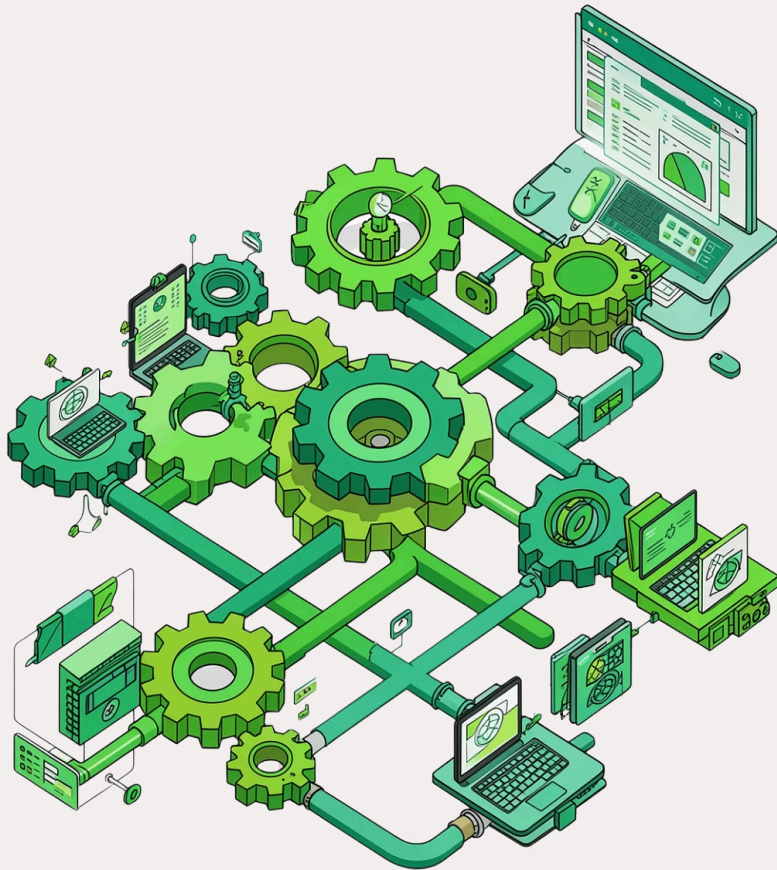
ISO 9001:2015 & ISO/IEC 27001 certified institution bridging academic learning with industry requirements through hands-on training.



Curriculum Overview

A rigorous 10-week journey through the full Machine Learning & Data Science stack.





Problem Statement

Unstructured E-Commerce Data

Product listings, prices, reviews, and availability are not readily available in structured form for analysis.

Manual Collection is Inefficient

Manual data gathering is time-consuming, error-prone, and cannot scale to support data science pipelines.

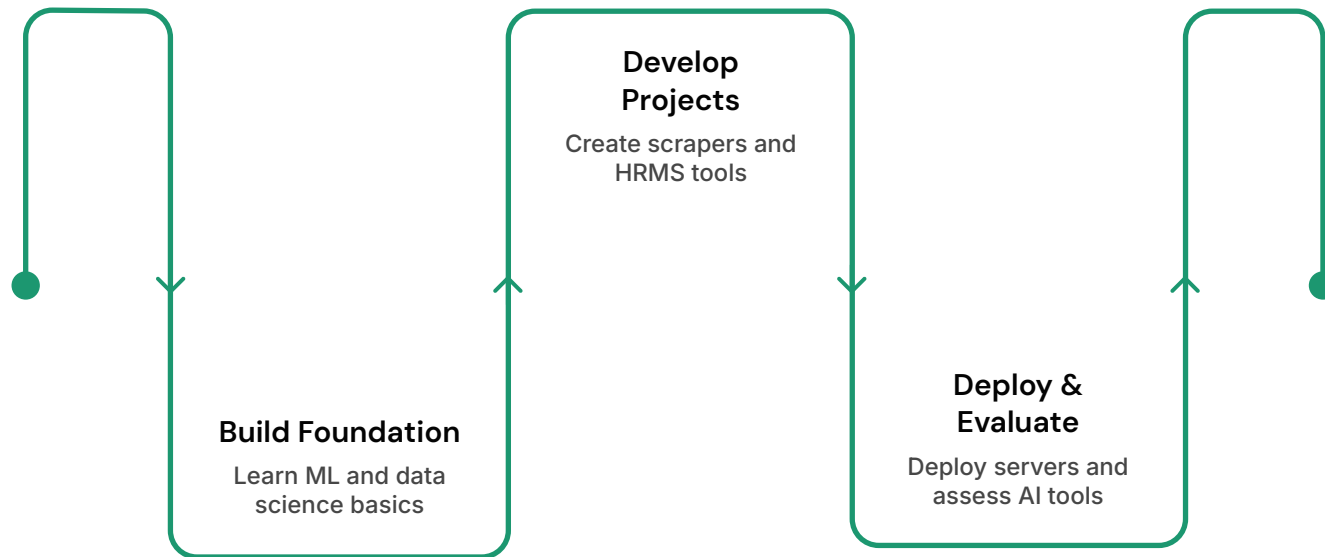
Internal Tool Gap

Organizations need customizable internal tools — such as HRMS systems — tailored to their workflows.

AI-Assisted Development

Opportunity to evaluate next-generation AI-assisted development tools in real-world project scenarios.

Project Objectives



Key Deliverables

- Build web scraper for Google Maps business listing data
- Design a generic, reusable web scraping framework
- Extend framework to 13 e-commerce & quick-commerce platforms
- Clone, customize, and prepare HRMS for deployment
- Perform comprehensive bug testing and quality checks
- Deploy completed projects on assigned servers
- Explore Antigravity IDE for AI-assisted development

Project 1: Google Maps Scraper

Objective

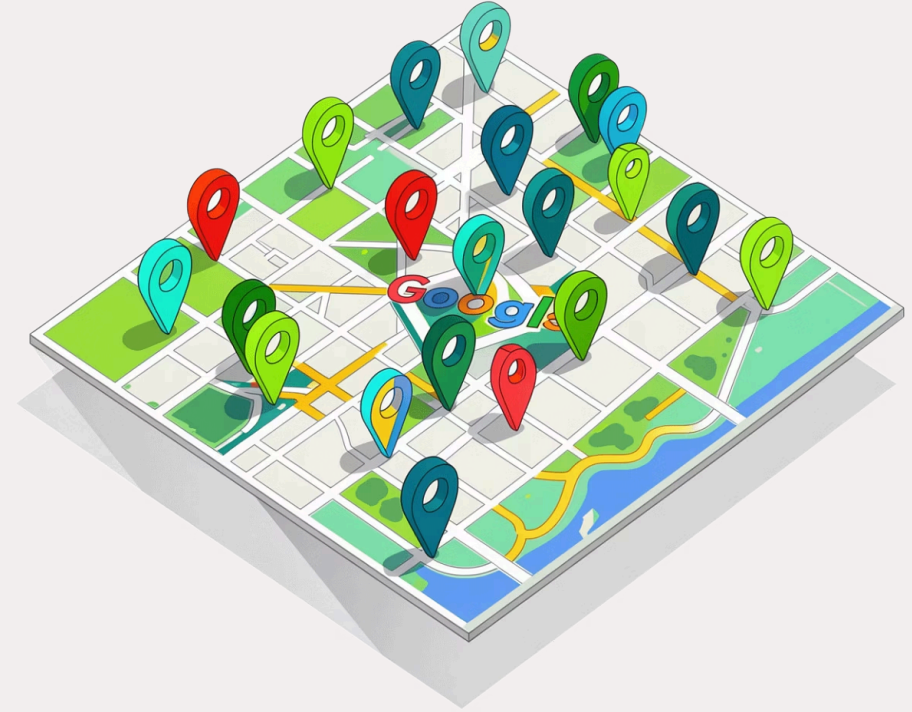
Extract structured business listing data — names, addresses, ratings, and contact details — directly from Google Maps.

Technology

Built with **Python** and the **Selenium** browser automation library for dynamic content rendering.

Outcome

Developed until server-side anti-scraping limitations were encountered; escalated to higher management for further evaluation.



Project 2: E-Commerce Web Scrapping Framework

Architecture

A **generic base module** extended to platform-specific scrapers — enabling rapid adaptation to each website's unique structure and anti-scraping measures.

Data Extracted

- Product names & prices
- Availability status
- Location details

Purpose

Provides the data foundation for the capstone project on **Customer Sentiment Analysis**.

10 Completed

Amazon · Flipkart ·
BigBasket · Zepto ·
Nykaa · Myntra · JioMart
· Purpille · Tira · Smytten

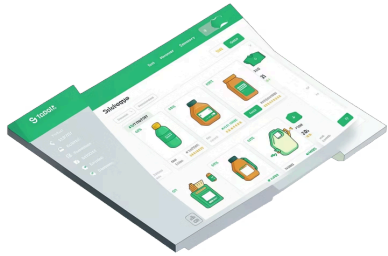
2 Pending

Blinkit (requires location
input) · Kult (website
maintenance)

Not Implemented

Alibaba

Project 2: Accompanying Website & Project 3: HRMS Customization



Accompanying Website

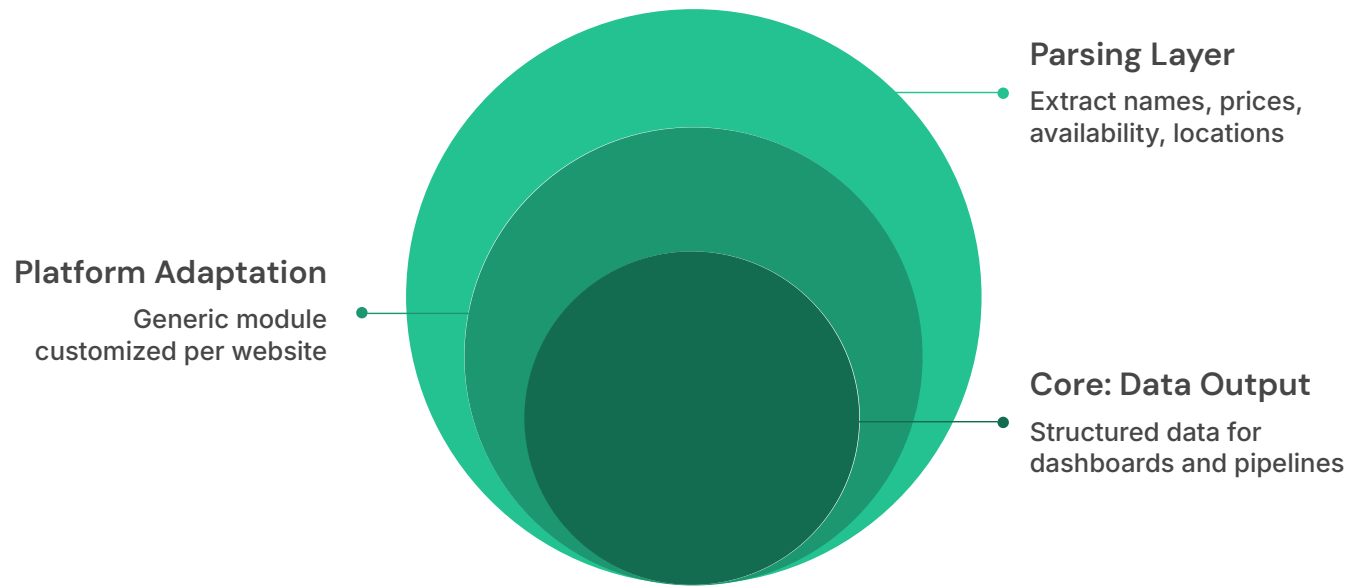
- Presents scraped multi-platform data in a user-friendly interface
- Front-end development fully completed
- Connected to web scraping framework outputs
- Deployment pending beyond internship tenure



HRMS Customization

- Cloned and customized an open-source HRMS from GitHub
- Client-server architecture for personal server deployment
- Incomplete due to early conclusion of internship tenure (resumption of college activities)

System Architecture



HRMS Architecture

Client-server model inherited from the open-source GitHub codebase, configured for deployment on a personal server.

Key Design Principles

- **Modularity:** Generic base extended per platform
- **Scalability:** Structured output feeds downstream pipelines
- **Resilience:** Handles anti-scraping measures per site

Technologies & Tools Used



Programming & Data

Python · NumPy · Pandas · Matplotlib · Seaborn



ML & NLP

Scikit-learn · NLP · Naive Bayes Classifier



Database & Deployment

SQL · Flask · Power BI Dashboards



Web Scraping & Dev Tools

Selenium · Antigravity IDE (AI-assisted) · GitHub

Key Achievements & Conclusion

10 / 13

E-commerce platform scrapers successfully completed and tested

10 Weeks

Full ML & Data Science curriculum completed at EduSkills Academy

3 Projects

Google Maps Scraper, E-Commerce Framework & HRMS — independently owned end-to-end

This internship provided hands-on experience designing and delivering real-world software projects — from automated web scraping pipelines to internal enterprise tools — solidifying both technical skills and professional engineering practices.



Methodology

Structured Learning

Week-by-week progression through curriculum modules, building depth progressively.

Iterative Development

Built generic solutions first, then extended and adapted them to specific platforms.

AI-Assisted Development

Antigravity IDE used across all practical projects to accelerate delivery.

Quality & Mentorship

Individual bug testing before completion, with regular guidance from an assigned mentor.



Key Achievements

10

Week Program

ML & Data Science certification
completed end-to-end

- **Google Maps Scraper**
Developed functional scraper
— escalated for senior review
due to complexity.

10

E-Commerce Platforms

Generic scraping framework
extended across all sites

- **Structured Data
Generation**
Produced clean e-commerce
datasets ready for sentiment
analysis pipelines.

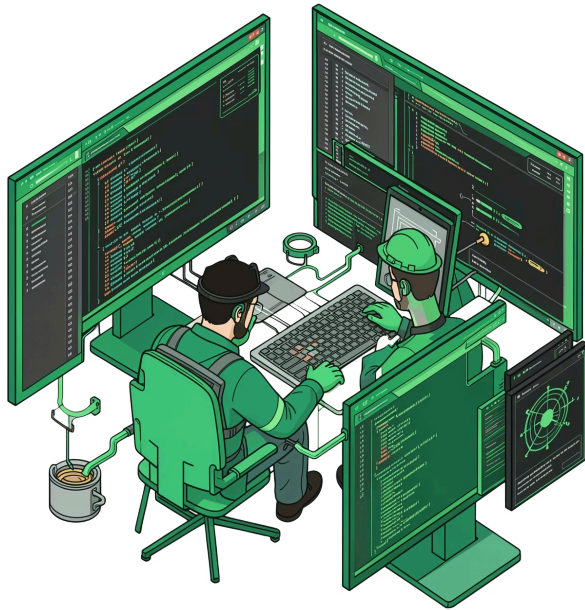
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Front-End Website

Built for multi-platform data
presentation

- **Independent Ownership**
Demonstrated full project
ownership from development
through deployment.

Challenges & Learning



Server-Side Limitations

Google Maps scraper encountered robust anti-scraping measures requiring escalation.

Platform Diversity

Each e-commerce site demanded custom adaptation of the base framework.

Dynamic Content

Selenium was required to handle JavaScript-heavy websites with dynamic rendering.



Positive Outcome: Gained lasting confidence in independent problem-solving and end-to-end deployment.

Skills Developed



ML & Data Science

Fundamentals of machine learning, model training, and deployment (MLOps).



Web Scraping

Browser automation with Selenium and scalable scraping frameworks.



AI-Assisted Dev

Practical experience with Antigravity IDE and AI development workflows.



Python & Data

Programming, data manipulation, and SQL database integration.



Data Visualization

Storytelling through structured data and front-end presentation.



Project Management

Independent ownership, problem-solving, and delivery with minimal supervision.

Conclusion

Program Completed

Successfully finished a comprehensive 10-week ML & Data Science internship.

Real-World Impact

Delivered practical software solutions addressing genuine industry needs.

Independence Proven

Demonstrated ability to manage and execute projects with minimal supervision.

Career Ready

Prepared to apply acquired skills in professional data science and software development roles.





Thank You

We welcome your questions and feedback.

Presenter

Raj Gupta

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